

Research Summary: AGN Reverberation Mapping Datasets for SURD Analysis

This document summarizes the datasets prepared for the Synergistic–Unique–Redundant Decomposition (SURD) causal information analysis of Active Galactic Nuclei (AGN).

1. Datasets Overview

We have prepared three primary datasets from well-known AGN monitoring campaigns, focusing on high-quality light curves suitable for measuring information flow between continuum and emission lines.

A. NGC 5548 (AGN Watch)

- **Campaign:** International AGN Watch (1988–2001)
- **Continuum:** Optical 5100 Å
- **Emission Line:** Integrated H β flux
- **Characteristics:** Long baseline (13 years), high signal-to-noise ratio.
- **Data Source:** [AGN Watch Archive](#)

B. NGC 5548 (STORM 1)

- **Campaign:** Space Telescope and Optical Reverberation Mapping (2014)
- **Continuum:** UV (1158 Å, 1367 Å, 1469 Å, 1745 Å)
- **Emission Lines:** Deblended Ly α , N V, Si IV, C IV, He II
- **Characteristics:** Exceptional daily cadence over 6 months, UV-focused.
- **Data Source:** [MAST HLSP STORM](#)

C. Mrk 817 (STORM 2)

- **Campaign:** STORM 2 (2020–2022)
- **Continuum:** UV 1180 Å
- **Emission Lines:** Ly α , C IV, He II
- **Characteristics:** Recent high-cadence monitoring of a highly variable AGN.
- **Data Source:** [MAST HLSP STORM 2](#)

2. File Structure and Formats

All datasets have been processed into clean CSV files with standardized column names:

Dataset	File Name	Columns	Units
NGC 5548 Watch	ngc5548_agnwatch_cont_clean.csv	mjd , flux , error	Flux: 10^{-15} erg/s/cm ² /Å
NGC 5548 Watch	ngc5548_agnwatch_hb_clean.csv	mjd , flux , error	Flux: 10^{-13} erg/s/cm ²
STORM 1	ngc5548_storm1_clean.csv	mjd , f_lya , e_f_lya , ...	Flux: 10^{-13} erg/s/cm ²
STORM 2	mrk817_storm2_continuum-1180_clean.csv	hjd-2400000 , flux , error	Flux: 10^{-13} erg/s/cm ² /Å

3. Velocity-Resolved Capability

For NGC 5548, we have downloaded the **H β Profiles (1989–1996)**. These consist of 247 high-quality spectra that can be integrated over specific velocity windows:

- **Blue Wing:** -6000 to -2000 km/s
- **Core:** -2000 to +2000 km/s
- **Red Wing:** +2000 to +6000 km/s

This allows for the analysis of information flow as a function of gas kinematics in the Broad Line Region (BLR).

4. Usage Instructions

1. **Load Data:** Use `pd.read_csv()` or the provided `agn_helpers.py`.
2. **Resample:** Use `prepare_surd_input` to interpolate continuum and line data onto a common grid.
3. **Standardize:** The helpers automatically apply zero-mean unit-variance scaling.
4. **SURD Analysis:** Feed the resulting `X = np.vstack([cont, line])` into your SURD implementation.